**UNIVERSITE LIBANAISE**
**Faculté des sciences - Section 3**

**الجامعة اللبنانية**
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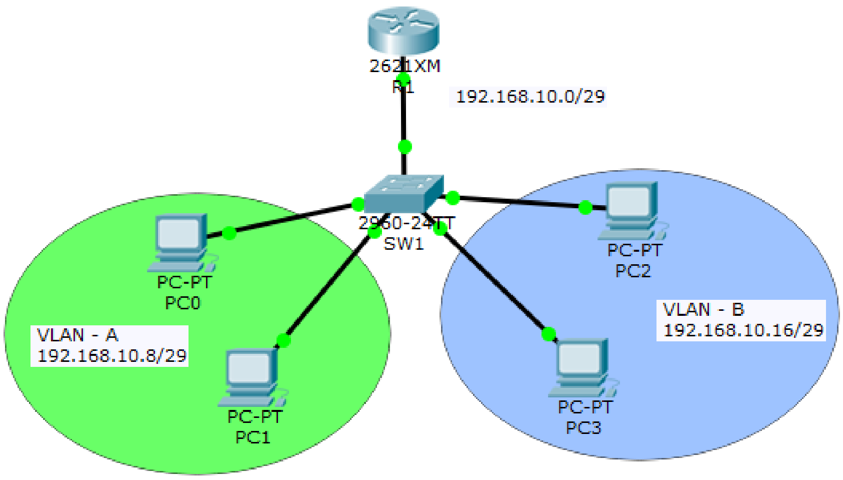
I3304 Computer Networks Administration and Security

# Lab2 – Configure Inter VLAN Routing on Cisco Router

Consider the following scenario




**Description:**

- Basically, on a VLAN, no host can communicate with hosts within other VLANs.
- It means only hosts that are members of the same VLAN can communicate with each other.
- So, if you want your VLANs hosts to be able to communicate with each other, you must configure inter-VLAN routing using a router or a layer 3 switch.
- In this Lab, we demonstrate the Inter-VLAN configuration.

## First step: Create 2 VLANs “VLAN A” and “VLAN B”

First of all, create two VLAN in the switch and named VLAN A and VLAN B by following the steps of Lab1. Then check them with “**show vlan**” command or “**do sh vlan**” command.

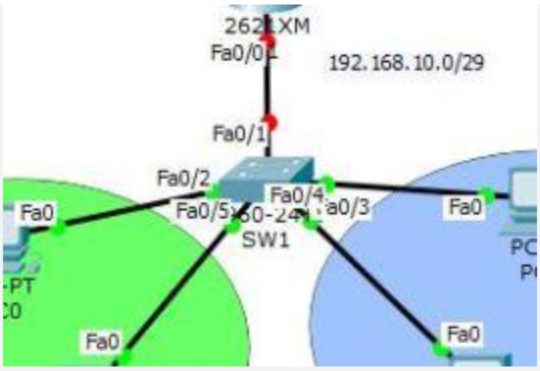
```
Switch#show vlan

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/1, Fa0/2, Fa0/3, Fa0/4
                                                Fa0/5, Fa0/6, Fa0/7, Fa0/8
                                                Fa0/9, Fa0/10, Fa0/11, Fa0/12
                                                Fa0/13, Fa0/14, Fa0/15, Fa0/16
                                                Fa0/17, Fa0/18, Fa0/19, Fa0/20
                                                Fa0/21, Fa0/22, Fa0/23, Fa0/24
                                                Gig0/1, Gig0/2
2    VLAN-A                           active
3    VLAN-B                           active
1002 fddi-default                     act/unsup
1003 token-ring-default               act/unsup
1004 fddinet-default                  act/unsup
1005 trnet-default                    act/unsup

VLAN Type  SAID       MTU   Parent RingNo BridgeNo Stp  BrdgMode Trans1 Trans2
```

Configure Inter-VLAN Routing

You see the result on the screenshot, the VLANs are ready for assigning switch ports to them.

## Step 2: Assigning Switch Ports for VLANs

In this section, the switch ports are divide and assign to VLANs. Before configuring Inter-VLAN routing, a host in a VLAN can only communicate within its own VLAN and not reach to other VLANs. So, let's configure it.



Assigning Switch Ports to VLANs

- Assign the port **FastEthernet 0/2** and **FastEthernet 0/5** to **VLAN 2** which named **VLAN-A**.
- Assign **FastEthernet 0/3** and **FastEthernet 0/4** to **VLAN-B**.

You will obtain:

```
Switch(config)#do sh vlan

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/1, Fa0/6, Fa0/7, Fa0/8
                                                Fa0/9, Fa0/10, Fa0/11, Fa0/12
                                                Fa0/13, Fa0/14, Fa0/15, Fa0/16
                                                Fa0/17, Fa0/18, Fa0/19, Fa0/20
                                                Fa0/21, Fa0/22, Fa0/23, Fa0/24
                                                Gig0/1, Gig0/2
2    VLAN-A                           active    Fa0/2, Fa0/5
3    VLAN-B                           active    Fa0/3, Fa0/4
1002 fddi-default                     act/unsup
1003 token-ring-default               act/unsup
1004 fddinet-default                  act/unsup
1005 trnet-default                    act/unsup

VLAN Type  SAID       MTU   Parent RingNo BridgeNo Stp  BrdgMode Trans1 Trans2
```

Assign Switch Ports to VLAN

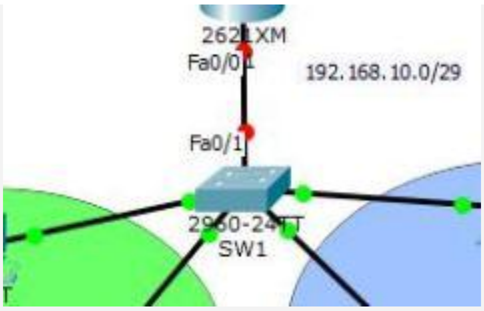
At this level,

- Only machines of the same VLAN can communicate together.
- The machines of the different VLANs can not communicate.

Ping the different machines and rest assured of the results, if errors are found, review the work and correct them.

## Step 3: Configure Trunking Ports on Switch

With the command "**switch port mode trunk**" you can configure trunking on the **FastEthernet 0/1** port of the **SW1**. The **VLAN Trunking Protocol** (VTP) let the VLANs transmit theirs traffics over a physical line simultaneously.



Configuring Trunk Ports

**1.** Just navigate to **FastEthernet 0/1** interface and type "**switchport mode trunk**", then press enter to enable trunking on **Fa0/1** interface line.

```
Switch(config)#interface fastEthernet 0/1
Switch(config-if)#switchport mode trunk
Switch(config-if)#
```

**2.** Now the VLANs can transmit traffic over the **FastEthernet 0/1** without any problems.

## Step 4 : Configure Inter-VLAN Routing on Cisco Router

Finally, the lab is ready to configure Inter-VLAN routing. If you test the PCs, they can ping with each other within a VLAN but not with other VLANs. So in order to communicate they need routing. Not network routing protocols such as **Static routing** or dynamic routing like **RIP**, and **OSPF**. Just need Inter-VLAN Routing which you simply configure according to below step by step Inter-VLAN routing guide.

**1.** Try to assign an IP address to the router and enable the interface you want to configure inter-VLAN routing.

VLAN Routing

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface fastEthernet 0/0
Router(config-if)#ip address 192.168.10.1 255.255.255.248
Router(config-if)#no shutdown
```

Good, the IP **192.168.10.1** with the subnet mask of **255.255.255.248** is assigned for physical **FastEthernet 0/0** interface.

**Note:** We need to have subinterface for each VLANs on the router. The subinterface is a virtual interface card that inter-VLAN doing routing using them.

**2.** Now try to create a subinterface for each VLAN with **interface** command and assign IP address from the different network for each VLAN. Do not forget that we use on this interface the IEEE 802.1q with the command "**encapsulation**". In this case, I have subnetted the **192.168.10.0/24** IP address to **3** networks.

```
Router(config)#interface fastEthernet 0/0.2
Router(config-subif)#
```

```
Router(config-subif)#encapsulation dot1Q 2
Router(config-subif)#ip address 192.168.10.9 255.255.255.248
Router(config-subif)#
```

**3.** See the result with “**do show ip interface brief**” from sub-interface area.

```
Router(config-subif)#do sh ip int bri
Interface              IP-Address      OK? Method Status                Protocol

FastEthernet0/0        192.168.10.1    YES manual up                    up

FastEthernet0/0.2      192.168.10.9    YES manual up                    up

FastEthernet0/1        unassigned      YES unset  administratively down down
Router(config-subif)#
```

Create Sub-Interface

The virtual sub-interface **FastEthernet0/0.2** has created and it has the 192.168.10.9 IP address. This sub-interface act as a default gateway for VLAN-A with an address of **192.168.10.8/29**.

**4.** Do the same to create a sub-interface for VLAN-B also.

```
Router(config)#interface fastEthernet 0/0.3
Router(config-subif)#ip address 192.168.10.17 255.255.255.248

% Configuring IP routing on a LAN subinterface is only allowed if that

subinterface is already configured as part of an IEEE 802.10, IEEE 802.1Q,
or ISL vLAN.

Router(config-subif)#encapsulation dot1Q 3
Router(config-subif)#ip address 192.168.10.17 255.255.255.248
Router(config-subif)#
```

**5.** Everything is fine, but you see the error with red color! It is because we forgot to set the **encapsulation dot1Q** command. Before assigning an IP address to a sub-interface, you should set IEEE 802.1q with **encapsulation** command.

Finally, all VLANs hosts can communicate with each other.